

Date: 2026-06-22

Request ID: 100048669

Hotplate		
Element(s)	Ag	
SOP#		
Cocktail		
sample(s)	weight (g)	volume (mL)
200128341_1	0.5042	200
200128341_2	0.5061	200
200128342_1	0.5035	200
200128342_2	0.5053	200

Hotplate		
Element(s)	Pd	
SOP#	Determination of Palladium in Catalyst with or without Ag present	
Cocktail		
sample(s)	weight (g)	volume (mL)
200128341_1	1.0065	200
200128341_2	1.0056	200
200128342_1	1.0042	200
200128342_2	1.0088	200

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128341	25.9833	28.0018	27.9891	0.63	0.9937

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128341_1	10/9	1.757	779.07	0.08
200128341_2	10/9	1.771	782.33	0.08

ag result: 780.70 ppm

ag result: 0.08 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Pd ICP analysis

sample	Dilution	conc. [mg/L]	Pd_ppm [dried]	%Pd [dried]
200128341_1	10/9	1.612	358.16	0.04
200128341_2	10/9	1.618	359.71	0.04

pd result: 358.93 ppm

pd result: 0.04 %

pd lot: *Manufacturer: Ricca, 841005E exp: 2027-11-30*

Note(s):

Date: 2026-06-22

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128342	27.1499	29.1655	29.1568	0.43	0.9957

Ag ICP analysis

sample	Dilution	conc. [mg/L]	Ag_ppm [dried]	%Ag [dried]
200128342_1	10/9	1.870	828.69	0.08
200128342_2	10/9	1.952	862.18	0.09

ag result: 845.43 ppm

ag result: 0.08 %

ag lot: *Manufacturer: Assurance, 28-111AGY exp: 2027-02-28*

Pd ICP analysis

sample	Dilution	conc. [mg/L]	Pd_ppm [dried]	%Pd [dried]
200128342_1	10/9	1.598	355.16	0.04
200128342_2	10/9	1.678	371.24	0.04

pd result: 363.20 ppm

pd result: 0.04 %

pd lot: *Manufacturer: Ricca, 841005E exp: 2027-11-30*

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}] * 10,000)$$

$$\%MO = [\text{oxide factor}] * \%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A] - [C-A]) / (B-A) * 100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1 - (\%LOI)] / 100)$$

$$\%M+ \text{ [dried]} = \%M+ / ([1 - (\%LOI)] / 100)$$

$$\%MO \text{ [dried]} = \%MO / ([1 - (\%LOI)] / 100)$$